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Association of compound exposure to heat wave and ozone pollution with cause-specific cardiopulmonary mortality: A space-time-stratified case-crossover study

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ABSTRACT

Heat wave and ozone (O_3) pollution events are increasingly co-occurring in the context of climate change, especially in China. However, the effects of compound exposure to both hazards on cause-specific cardiopulmonary mortality remain largely unclear. In this study, we conducted a space-time-stratified case-crossover study of 550,634 township-level cardiopulmonary deaths in Anhui Province, China, from 2013 to 2021 to systematically evaluate the associations of exposure to heat wave and O_3 pollution with multi-cause cardiopulmonary mortality and quantify their interactive effects Under various definitions, the odds ratios (ORs) of cardiopulmonary mortality associated with heat wave-only events, O_3 pollution-only events, and compound events exposure ranged from 1.15 (95 % CI: 1.14–1.16) to 1.39 (95 % CI: 1.37–1.41), 1.02 (95 % CI: 1.01–1.03) to 1.04 (95 % CI: 1.03–1.04), and 1.22 (95 % CI: 1.20–1.23) to 1.75 (95 % CI: 1.62–1.88), respectively. Notably, more significant synergistic effects of heat wave and O_3 pollution exposure at lag 1 day were observed on mortality risk from hypertensive heart disease, ischemic heart disease, acute myocardial infarction, hemorrhagic stroke, and chronic obstructive pulmonary disease. Females, older adults, and residents living in rural-dominated mixed regions experienced higher compound exposures. During the study period, the estimated number of excess cardiopulmonary deaths attributable to compound events exposure under various definitions ranged from 243 (95 % CI: 201–285) to 3227 (95 % CI: 2974–3480). More than 90 % of these excess deaths occurred after 2017 and were primarily concentrated in densely populated northern and central-eastern townships. These findings underscore that coordinated governance of air pollution and climate change could maximize reductions in the cardiopulmonary mortality burden.

1. Introduction

Climate change is driving a continuous rise in global temperatures, leading to rapid increases in the frequency, intensity, and duration of heat wave events (IPCC, 2023; Romanello et al., 2024; Jha et al., 2025). Concurrently, ambient ozone (O_3) pollution has become an increasingly prominent concern, with over 90 % of the global population living in areas where O_3 concentrations exceed the World Health Organization's guideline limits of 100 μg/m^3 (WHO, 2023; Domingo et al., 2024; Sun et al., 2024). Due to their shared meteorological drivers and conditions (Meehl et al., 2018; Ji et al., 2024), heat wave and O_3 pollution frequently co-occur during warm seasons, readily forming compound events that pose greater health threats (Zong et al., 2022; Wu et al., 2023; Gao et al., 2025). However, little is known regarding the joint health effects of heat wave and O_3 pollution. Projection studies further indicate that the annual mean number of such compound event days will increase by 5.5–34.6 days in the 2080s, with expanding impacted areas, potentially causing immeasurable losses of life and livelihoods

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worldwide (Ban et al., 2022; Liu et al., 2024; Fang et al., 2025). Therefore, it is both crucial and urgent to explore the effects of compound exposure to these two hazards.

Exposure to high temperature and O_3 pollution can induce oxidative stress, systemic inflammation, autonomic nervous system dysfunction, and promote thrombosis, all of which together exacerbate pre-existing conditions and trigger severe cardiopulmonary events (Wang et al., 2022, 2025c; Renaghan et al., 2024). Over the past decade, the adverse effects of heat wave and O_3 pollution exposure on cardiopulmonary mortality have been extensively documented (Yan et al., 2023; Zhang et al., 2024; Tran et al., 2025). Given the consistent evidence and potential shared biological mechanisms, several recent studies have linked exposure to concurrent heat wave and O_3 pollution with cardiopulmonary outcomes. For example, in California, extreme heat events and O_3 pollution exposure synergistically increased respiratory hospitalizations among certain community residents, though the synergistic effect was not significant at the state level (Schwarz et al., 2021). Two case-crossover studies in Jiangsu, China, indicated that the compound events were consistently associated with higher risks of all-cause cardiovascular and ischemic stroke mortality than the heatwave-only and high O_3 pollution-only events (Xu et al., 2023a; Wang et al., 2025c). However, existing population-based studies have focused on all-cause cardiopulmonary or specific cerebrovascular outcomes and have not yet systematically elucidated the joint effects of heat wave and O_3 pollution on specific cardiopulmonary subtypes. Although a time-series study across 250 Chinese counties reported associations between compound exposure to heat wave and O_3 pollution and multi-cause cardiopulmonary deaths (Du et al., 2024), substantial gaps remain in confounding control and assessment accuracy. Notably, current studies generally define heat wave events as days with daily mean temperature exceeding the 90th percentile; however, the independent and joint effects related to heat wave events defined by lower thresholds and longer durations are often overlooked, potentially creating blind spots in health protection efforts. Additionally, the spatial distribution of health outcomes and exposure factors often exhibit complex spatiotemporal characteristics (Sun et al., 2023; Jiang et al., 2024). Currently, population-based compound exposure studies are somewhat insufficient in controlling for spatial confounding factors. Therefore, the acute impacts of compound exposure to heat wave and O_3 pollution on multi-cause cardiopulmonary mortality still lack refined, systematic, and large-scale epidemiological evidence, limiting in-depth interpretation of sensitive cardiopulmonary mortality risks.

To this end, we conducted a township-level space-time-stratified case-crossover study based on over 0.5 million cardiopulmonary deaths in Anhui Province, China, during 2013–2021. Our primary aim was to comprehensively evaluate the association of exposure to heat wave and O_3 pollution with cause-specific cardiopulmonary mortality, quantify their interactive effects on various cardiopulmonary mortality outcomes, and further identify sensitive diseases and vulnerable populations. This study will facilitate the rational allocation of medical resources and provide novel approaches and strategies to support joint warning systems and preventive measures.

2. Material and methods

2.1. Study area

The study was conducted in Anhui, an eastern province of China (114°54'–119°37' E, 29°41'–34°38' N). Anhui Province is situated in the transitional zone between the warm temperate and subtropical regions, characterized by both a warm temperate semi-humid monsoon climate and a subtropical humid monsoon climate, with significant differences between the north and south. It covers a total area of 140,100 square kilometers and has a resident population of approximately 61.23 million in 2024. Among the residents, females accounted for 49.03 %, rural residents for 41.67 %, and individuals aged over 65 years for 15.01 % (https://tjj.ah.gov.cn/). Since 2013, heat wave events in the region have become increasingly frequent, and near-surface O_3 pollution levels have shown an overall upward trend (Ding and Chen, 2024; Wang et al., 2025a). Moreover, Anhui is among the regions in China with the weakest control and prevention of cardiovascular diseases (Mao et al., 2022). These conditions make Anhui a typical study area for investigating the associations between heat wave, O_3 pollution, and cause-specific cardiopulmonary mortality.

2.2. Outcomes

Mortality from cardiopulmonary as the underlying cause of death was the study outcome of interest. Daily death records for the warm seasons (May 1 to September 30) from 2013 to 2021 were obtained from the mortality surveillance system of Anhui Provincial Center for Disease Control and Prevention. Mortality data were classified according to the International Statistical Classification of Diseases and Related Health Problems, 10th Revision (ICD-10), including circulatory diseases (codes I00–I99), cardiovascular diseases (codes I00–I59), cerebrovascular diseases (codes I60–I69), and respiratory diseases (codes J00–J99). In addition, nine specific cardiopulmonary mortality categories were considered: hypertensive heart disease (codes I10–I15), myocardial infarction (codes I21–I23), acute myocardial infarction (codes I21–I22), ischemic heart disease (codes I20–I25), chronic ischemic heart disease (code I25), stroke (codes I60–I64), hemorrhagic stroke (codes I60–I61), ischemic stroke (code I63), and chronic obstructive pulmonary disease (codes J41–J44). Since township represents the smallest administrative unit in the death registration reporting system, we selected it as the spatial units for analysis. During the study period, the death records from 122 townships were incomplete. After excluding subjects with missing detailed registration information from these townships ($n = 4{,}123$, 0.74 %), a total of 550,634 death cases from the remaining 1390 townships were included in the analysis. To further identify vulnerable groups, township-level daily mortality data for these causes were classified by age (<65 and $\geq$ 65 years), sex (male and female), and region (UR, MU, MR and RU; see Supporting Materials Text S1 for Detailed classification criteria and formulas).

2.3. Study design

A space-time-stratified case-crossover design was employed to evaluate the association of short-term exposure to heat wave and O_3 pollution with cause-specific cardiopulmonary mortality. This method uses each subject as their own control, matching case days and multiple control days by the fixed time and space stratum. By comparing exposures on case day and control days, the relationship between environmental exposures and outcomes can be determined. For each case, the date of death was the case day, and 3 or 4 control days were matched by day of the week within the same month, year, and township. This design inherently controls for the effects of day of the week, seasonality, long-term trend, spatial variation, and individual characteristics, thereby more accurately revealing the independent and joint effects of heat wave and O_3 pollution on cause-specific cardiopulmonary mortality.

2.4. Exposure assessment

Daily maximum 8-h average O_3 concentration data from 2013 to 2021 were obtained from China High Air Pollutants (CHAP) dataset (spatial resolution: 1km × 1 km). To examine the confounding effects of co-pollutants, daily $PM_{2.5}$ and NO_2 concentration data were also extracted from CHAP. From the European Centre for Medium-Range Weather Forecasts Reanalysis v5-Land (ERA5-Land) dataset, we obtained daily average air temperature (°C), dew point temperature (°C), and wind speed (m/s) over the same period (spatial resolution of 0.1° × 0.1°). Relative humidity (%) for each grid cell was then calculated using the average air temperature and dew point temperature. To maintain consistency, all these meteorological variables were resampled to a spatial resolution of 1 km × 1 km using a bilinear interpolation (Wang et al., 2025b). Compared to air temperature alone, apparent temperature integrates multiple meteorological factors, including not only temperature but also humidity, wind speed, and radiation, which collectively influence the human body's thermal perception and thermoregulation mechanisms.Specifically, human physiological responses to heat depend on ambient temperature as well as humidity, which affects sweat evaporation; wind speed, which influences convective heat loss; and radiant heat exchange. Therefore, apparent temperature provides a more comprehensive representation of actual thermal stress. By capturing these joint effects, this comprehensive metric enables more precise identification of physiological critical thresholds associated with heat stress, thereby offering a more accurate assessment of temperature's impact on population health (Schwingshackl et al., 2021; Xu et al., 2023). Daily apparent temperature (°C) was calculated employing the Steadman RG (1984) method; the formula for *AT* is as follows:

$$AT = T + 0.33 \times \mathrm{e} - 0.70 \times ws - 4.00 \tag{1}$$

$$\mathrm{e} = \frac{rh}{100} \times 6.105 \times \exp\left(\frac{17.27 \times T}{237.7 + T}\right) \tag{2}$$

where, *AT* represents the apparent temperature (°C), *T* denotes the mean air temperature (°C), *e* stands for water vapor pressure (hPa), *ws* indicates wind speed (m/s), and *rh* signifies relative humidity (%).

Considering differences in climatic conditions and population heat acclimatization, we applied combinations of multiple apparent temperature thresholds and durations to define heat wave (Ban et al., 2022; Xu et al., 2023). For each township, heat wave days were defined as at least 2, 3, 4, or 5 consecutive days with daily apparent temperature equal to or exceeding the 85th, 87.5th, 90th, 92.5th, and 95th percentiles of the study period. According to this approach, we built 20 definitions for heat wave at the township level. O_3 pollution days were categorized into low- (≥ the 75th percentile), medium- (≥160 μg/m^3), and high-level (≥ the 95th percentile). The medium-level O_3 pollution threshold was determined based on the secondary concentration limit for O_3 specified in the Ambient Air Quality Standard of China (CAAQ; GB3095-2012). By combining the definitions of heat wave and O_3 pollution, a total of 60 township-level gradient compound exposure scenarios were constructed (Table S1). We then matched township-level estimates of heat wave and O_3 exposure to each subject on case day and control days, based on their residential address and date of death.

2.5. Statistical analysis

We used conditional logistic regression models to evaluate the associations of exposure to heat wave and O_3 pollution with cause-specific cardiopulmonary mortality. In the model, event day was assigned as the independent binary indicator variable (occurrence or non-occurrence), and the indicator of case-control status (1 or 0) served as the outcome measure (Wu et al., 2021). Since this study design inherently controls for individual-level covariates such as age and sex, as well as temporal and spatial confounders including seasonality and spatial heterogeneity, the primary models included only binary indicator variables representing the three extreme events. In light of recent evidence highlighting the joint health effects of $PM_{2.5}$, NO_2, and relative humidity when combined with heat wave and O_3 pollution (Chen et al., 2023; Xu et al., 2023a, 2023b), sensitivity analyses were performed to further adjust for these potential confounders and to assess their impact on the results. We also examined the lag effects of the three days before death for different diseases (Du et al., 2024), with 4 single lag days analyzed (same day, lag 1, lag 2, lag 3). We then estimated the overall odds ratio (OR) by selecting the maximum points estimate among the 4 lag days under various definitions (Guo et al., 2023).

The relative excess odds due to interaction (REOI), proportion attributable to interaction (AP), and synergy index (S) were utilized to further quantify the relative difference between the joint effect of concurrent heat wave and O_3 pollution and the sum of their individual effects. Specifically, REOI measures the absolute excess risk due to interaction between heat wave and O_3 pollution, AP estimates the proportion of risk among individuals exposed to both hazards that is attributable to their interaction, and S assesses whether the joint effect exceeds the sum of individual effects, reflecting the strength of synergism (Knol et al., 2011). However, each metric has limitations. REOI and AP depend on correct model specification and may be affected by unmeasured confounding, while S requires sufficient statistical power and precise risk estimates for reliable interpretation (Knolm and Vanderweele, 2012). To overcome these limitations and capture different facets of interaction, we combined all three indices to provide a more comprehensive and robust assessment of the interaction effects. These three indicators are calculated using the following formulas:

$$\begin{aligned} \mathrm{REOI} &= (\mathrm{OR}_{11} - 1) - (\mathrm{OR}_{10} - 1) - (\mathrm{OR}_{01} - 1) \\ &= \mathrm{OR}_{11} - \mathrm{OR}_{10} - \mathrm{OR}_{01} + 1 \end{aligned} \tag{3}$$

$$AP = \frac{REOI}{OR_{11}} \tag{4}$$

$$S = \frac{OR_{11-1}}{(OR_{1}0 - 1) + (OR_{01-1)}} \tag{5}$$

where OR_{11}, OR_{10}, and OR_{01} represent the OR for cause-specific cardiopulmonary mortality associated with compound exposure days, heat wave-only days, and O_3 pollution-days relative to the reference days ($OR_{00} = 1$), respectively. REOI >0 (with 95 % CI excluding 0), AP > 0 (with 95 % CI excluding 0), and S > 1 (with 95 % CI excluding 1) indicate that the joint effect of heat wave and O_3 pollution on cause-specific cardiopulmonary mortality exceeds the sum of their individual effects (i.e., synergistic effect); otherwise, no additive interaction exists. A delta method was used to calculate the corresponding 95 % CIs for REOI, AP and S.

We further conducted stratified analyses by age (<65 and ≥65), sex (male and female), region (UR, MU, MR and RU) to assess cardiopulmonary mortality risks associated with the three extreme events. A two-sample Z-test was used to determine whether the differences in effect estimates between subgroups were statistically significant:

$$z = \frac{\beta_1 - \beta_2}{\sqrt{SE_1^2 + SE_2^2}} \tag{6}$$

where $\beta 1$ and $\beta 2$ represent the effect estimates for two subgroups (e.g., males and females), while SE_1 and SE_2 denote their corresponding standard errors.

Using the estimated associations, we calculated the excess cardiopulmonary deaths attributable to exposure to heat wave-only, O_3 pollution-only, and compound events for each township from 2013 to 2021 under various definitions, respectively. The calculation formula is as follows:

$$ED_{\mathrm{i,t,j}} = Pop_{i,t} \times \mathrm{Age}_{i,t,j} \times Mort_{i,t,j} \times \left(OR_{i,j} - 1\right) \times D_{\mathrm{i,t}} \tag{7}$$

$ED_{i,t,j}$ represents the excess deaths attributed to exposure to the three extreme events in the *j* age group of township *i* in year *t*; $Pop_{i,t}$ is the total population of township *i* in year *t*; $Age_{i,t,j}$ denotes the proportion of the *j* age group in township *i* in year *t*. The initial age-specific population data were derived from the WorldPop dataset developed by the University of Southampton (https://hub.worldpop.org/). We then adjusted the WorldPop population data for each year using township-level age data from the Seventh National Population Census in 2020. $Mort_{i,t,j}$ denotes the expected daily mortality rate for the *j* age group in township *i* in year *t*, which is calculated based on the Chinese standard population (sourced from the Global Burden of Disease, GBD) and the all-cause deaths on non-event days across 1390 townships annually (Zhang et al., 2021). $OR_{i,j}$ is the odds ratio of cardiopulmonary mortality for the j age group in township i associated with the three extreme events exposure. $D_{i,t}$ represents the number of exposure days to the relevant extreme event in township i during year t.

2.6. Sensitivity analysis

To check the robustness of our results, several sensitivity analyses were performed with respect to 1) defining heat wave events using daily mean air temperature instead of apparent temperature, 2) adjusting for relative humidity exposure in the past 3 days as a natural cubic spline with 3 df in the main model, 3) adjusting for NO_2 and $PM_{2.5}$ exposure as a natural cubic spline with 3 df in the main model to evaluate the potential impacts of co-pollutants on the results, 4) accounting for spatial autocorrelation by calculating cluster-robust standard errors with counties as the clustering units, and 5) restricting to subjects died before 2020 to mitigate potential confounding effects associated with the COVID-19 pandemic.

3. Results

3.1. Descriptive statistics

Table 1 summarizes the demographic characteristics of 550,634 cardiopulmonary deaths. During the warm seasons from 2013 to 2021 in 1390 Anhui townships, the total deaths counts were 474,639, 235,461, 237,468, and 75,995 for circulatory, cardiovascular, cerebrovascular, and respiratory mortality, respectively (Table 1). Of these subjects, the proportions of females, over 65 years, married, population with below primary education, and farmers were 46.10 %, 86.02 %, 61.70 %, 95.73 %, and 81.61 %, respectively (Table S2). Mean exposure to O_3, $PM_{2.5}$, and NO_2 across all study townships was 113.59 ± 38.12 μg/m^3, 34.77 ± 18.91 μg/m^3, and 21.48 ± 7.93 μg/m^3, respectively, whereas the average daily apparent temperature and relative humidity were 27.59 ± 5.20 °C and 75.86 % ± 11.76 % (Table S3). During the study period, the mean apparent temperature threshold of each township across Anhui was 33.31 °C, 33.82 °C, 34.38 °C, 35.03 °C, and 35.81 °C for the 85th, 87.5th, 90th, 92.5th, and 95th percentiles, respectively, while the 75th and 95th percentiles of daily maximum 8-h concentrations of O_3 were 139 μg/m^3 and 185 μg/m^3, respectively (Table S1).

3.2. Exposure distribution

From the most lenient definition (HW85_2d-OZ75th) to the most stringent one (HW95_5d-OZ95th), the number of township-days exposure to heat wave-only, O_3 pollution-only, and compound events in Anhui decreased from 258,349 (12.48 %), 458,369 (22.15 %), and 48,431 (2.34 %) to 54,382 (0.58 %), 80,110 (3.87 %), and 1128 (0.05 %), respectively (Table S4). Overall, the number of township-days related to the three types of extreme events gradually declined with increasing thresholds and duration. Moreover, the spatial distribution of exposure days at the township level was closely associated with their exposure levels. Townships with a higher number of low-intensity heat wave days were concentrated in the northern regions, whereas medium- and high-intensity heat wave days were widely distributed across central and southern townships (Fig. 1a, d, and g). The southern and northern townships experienced more frequent low- and high-level O_3 pollution (Fig. 1b and h), whereas medium-level O_3 pollution showed a spatial gradient that decreased from north to south (Fig. 1e), thereby demonstrating a clear and notable spatial heterogeneity in O_3 pollution distribution across the townships. Additionally, townships experienced compound exposure to medium-low level of heat wave and O_3 pollution were mainly observed in the northern and central regions (Fig. 1c and f). Under high-level heat wave and O_3 pollution exposure, townships with more compound exposure days were concentrated in the densely populated areas of central-eastern Anhui (Fig. 1i).

3.3. Risk associations

Based on the lag–response patterns of both the independent and joint effects of heat wave and O_3 pollution events, a lag period of 0–3 days was selected (Figs. S1–S2; Fig. 3B). We observed that exposure to heat wave-only and O_3 pollution-only events significantly increased the risk of cause-specific cardiopulmonary mortality, with the mortality risk being more pronounced at lag 1 day (Figs. S1–S2). Under various definitions, the ORs for cardiopulmonary mortality associated with exposure to heat wave-only and O_3 pollution-only events ranged from 1.15 (95 % CI, 1.14–1.16) to 1.39 (95 % CI, 1.37–1.41) and from 1.02 (95 % CI, 1.01–1.03) to 1.04 (95 % CI, 1.03–1.04), respectively (Table S5). This association generally strengthened with increasing thresholds and durations, although for high-level O_3 pollution events, the association with circulatory mortality risk was observed to be reduced (Table S8). In contrast, high-level O_3 pollution events substantially impacted respiratory mortality more than circulatory mortality, with chronic obstructive pulmonary disease showing the highest susceptibility (Fig. S1;

Table 1
Characteristics of cause-specific cardiopulmonary mortality in Anhui Province, China during the warm seasons from 2013 to 2021.

Cause	ICD-10 codes	N (%)	Mean (SD)	Median (IQR)
Cardiopulmonary	I00–I99	550,634	61,182 (17,635)	62,546 (30,250)
Male	J00–J99	296,791 (53.90)	32,977 (9261)	33,569 (15,754)
Female		253,843 (46.10)	28,205 (8375)	28,977 (14,496)
0–64 years		76,974 (13.98)	8553 (1544)	8634 (1969)
Over 65 years		473,660 (86.02)	52,629 (16,811)	54,349 (27,338)
Circulatory	I00–I99	474,639 (86.20)	52,738 (16,384)	53,143 (28,257)
Cardiovascular	I00–I59	235,461 (42.76)	26,162 (8000)	27,053 (13,214)
Hypertensive Heart	I10–I15	47,306 (8.59)	5256 (1310)	5866 (2011)
Acute Myocardial Infarction	I21–I22	86,018 (15.62)	9558 (3395)	9097 (6131)
Myocardial Infarction	I21–I23	86,053 (15.63)	9561 (3395)	9102 (6132)
Ischemic Heart	I20–I25	160,948 (29.23)	17,883 (7430)	17,087 (12,254)
Chronic Ischemic Heart	I25	73,210 (13.30)	8134 (4029)	7795 (6068)
Cerebrovascular	I60–I69	237,468 (43.13)	26,385 (8335)	25,911 (14,952)
Cerebral Stroke	I60–I64	192,889 (35.03)	21,432 (6132)	21,549 (11,258)
Hemorrhagic Stroke	I60–I61	102,793 (18.67)	11,421 (2550)	11,799 (4107)
Ischemic Stroke	I63	83,466 (15.16)	9274 (3555)	8955 (6219)
Respiratory	J00–J99	75,995 (13.80)	8444 (1396)	9104 (1445)
Chronic Obstructive Pulmonary	J41–J44	56,431 (10.25)	6270 (1414)	6719 (1543)

Abbreviations: SD, standard deviation; IQR, interquartile range.
Note: The mean and median values presented in the table are aggregated by year.

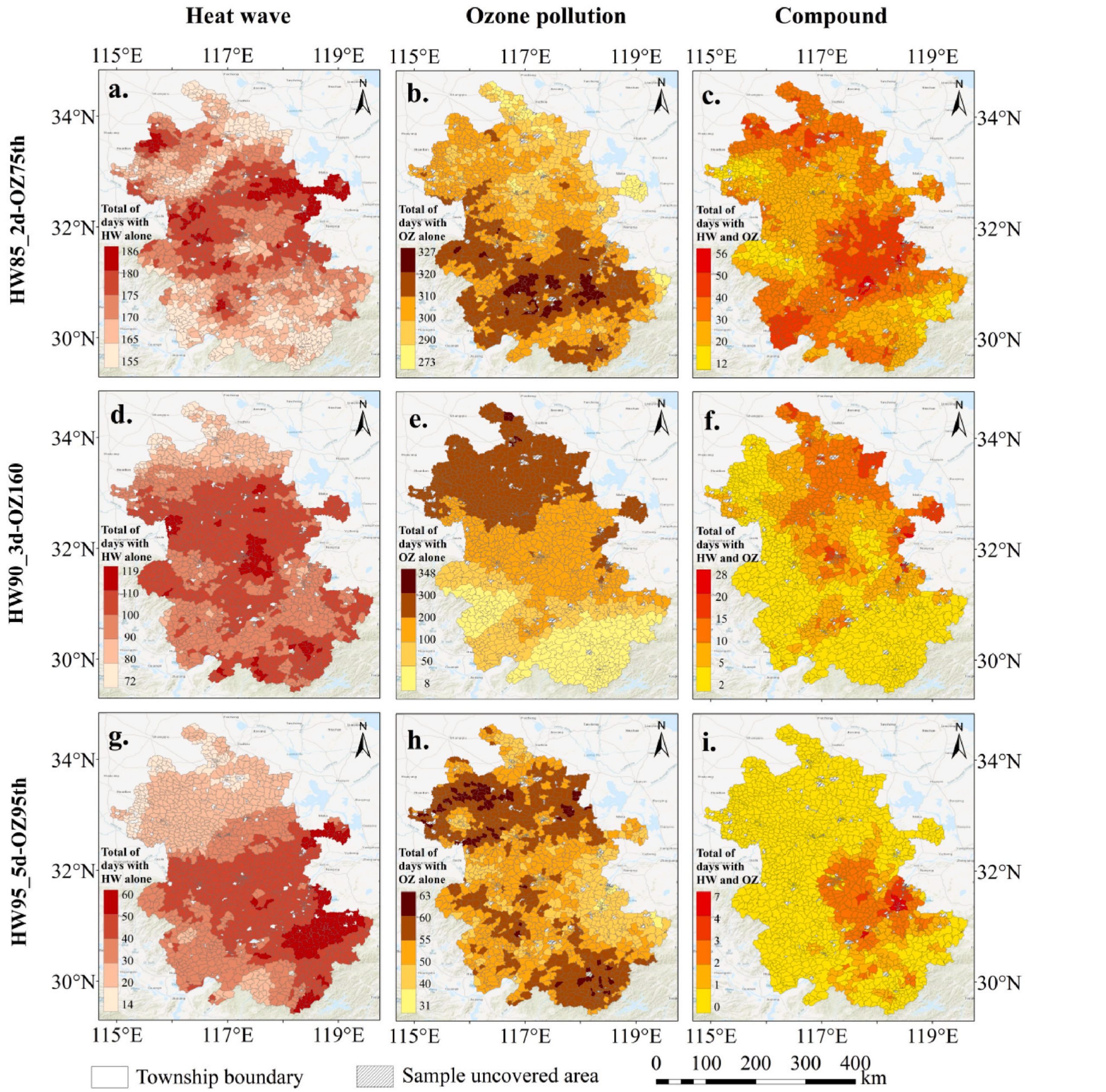

Fig. 1. Spatial distribution of the total number of expose days in 1390 Anhui townships during the warm seasons from 2013 to 2021 under various definitions.

Table S5). We also found a more pronounced impact of heat wave events on circulatory mortality. Notably, hypertensive heart disease, ischemic stroke, and chronic ischemic heart disease exhibited higher susceptibility to heat wave exposure, with ORs across different exposure levels ranged from 1.22 (95 % CI, 1.19–1.25) to 1.55 (95 % CI, 1.48–1.62), 1.20 (95 % CI, 1.17–1.22) to 1.48 (95 % CI, 1.42–1.54), and 1.17 (95 % CI, 1.15–1.20) to 1.48 (95 % CI, 1.42–1.54), respectively (Fig. 2).

The compound events were consistently associated with higher cardiopulmonary mortality risk than the heat wave-only and O_3 pollution-only events, with effect estimates generally being the strongest at lag 1 day (Fig. 3A; Fig. S3). Under 60 definitions, the ORs of cardiopulmonary mortality associated with compound events exposure ranged from 1.22 (95 % CI, 1.20–1.23) to 1.75 (95 % CI, 1.62–1.88), and were significantly positively correlated with both heat wave intensity and O_3 pollution levels (Table S8). This pattern remained consistent across cause-specific cardiopulmonary mortality. Notably, compound exposure to heat wave and O_3 pollution was associated with more pronounced increases in mortality risk for hypertensive heart disease, acute myocardial infarction, ischemic heart disease, ischemic stroke, and chronic obstructive pulmonary disease (Fig. 2; Tables S5 and S6). The corresponding OR ranges under various definitions were 1.33 (95 % CI, 1.27–1.39) to 1.88 (95 % CI, 1.62–2.15), 1.23 (95 % CI, 1.19–1.27) to 1.89 (95 % CI, 1.48–2.30), 1.21 (95 % CI, 1.18–1.24) to 1.88 (95 % CI, 1.56–2.27), 1.25 (95 % CI, 1.20–1.29) to 1.67 (95 % CI, 1.35–2.00), and 1.22 (95 % CI, 1.17–1.27) to 2.00 (95 % CI, 1.54–2.46), respectively.

3.4. *Joint effects*

REOI, AP, and S further revealed the joint effects of heat wave and O_3 pollution on cause-specific cardiopulmonary mortality. Since effect estimates were generally the strongest at lag 1 day, we used the lag-1 estimates to calculate the additive interaction. The findings indicated significant synergistic effects of heat wave and O_3 exposure on cardiopulmonary mortality (except for HW87.5_4d-OZ75th, HW87.5_5d-OZ75th, and HW90_2d-OZ75th). Across 57 definitions, the ranges of REOI, AP, and S were 0.03 (95 % CI, 0.02–0.05) to 0.39 (95 % CI,

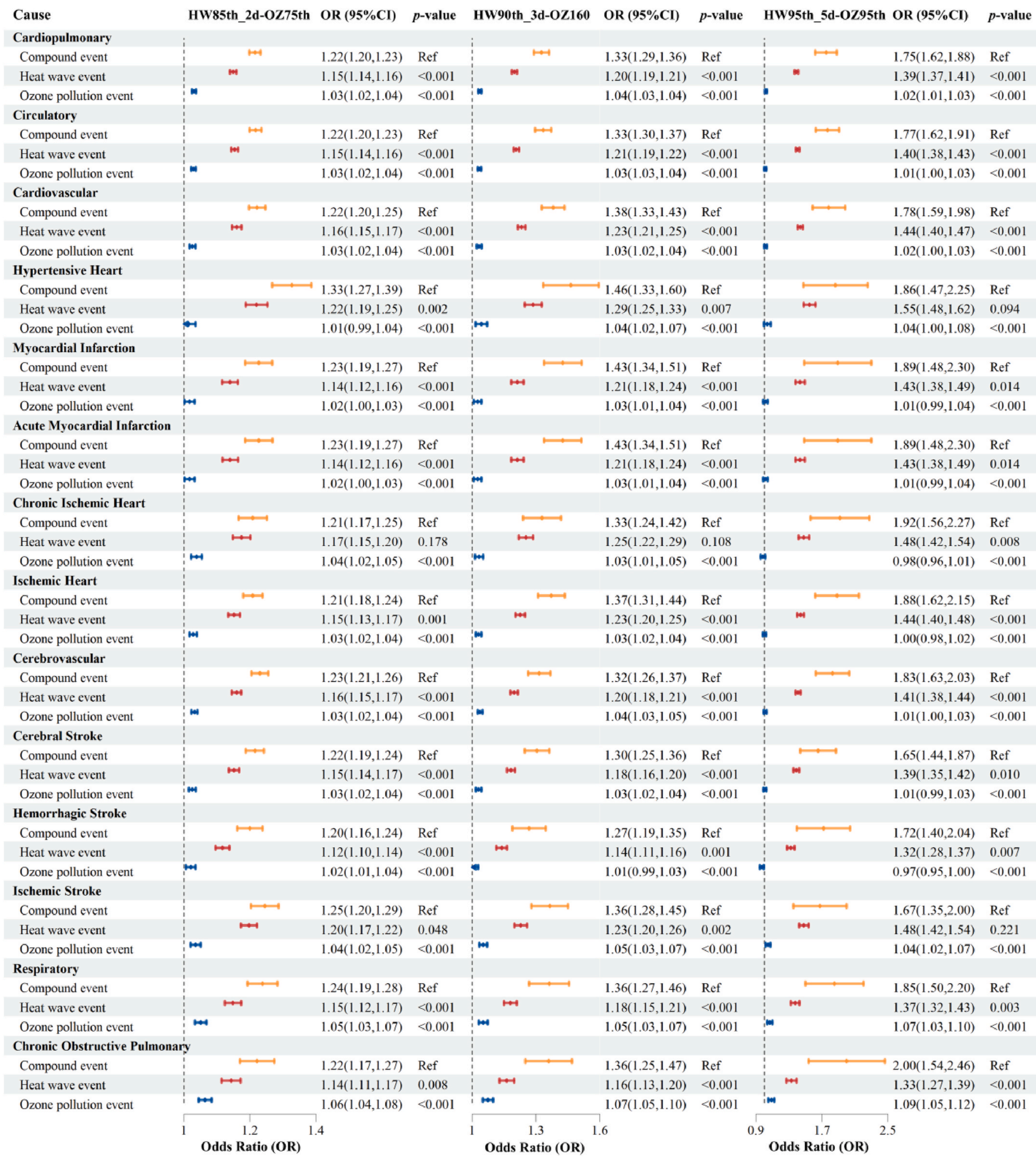

Fig. 2. Cause-specific cardiopulmonary mortality risk at lag 1 day for heat wave-only, and O_3 pollution-only and compound events under various definitions. In the forest plots, colors indicate event types: red denotes heat wave-only events, blue denotes O_3 pollution-only events, and yellow denotes compound events. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

0.26–0.51), 0.02 (95 % CI, 0.01–0.04) to 0.23 (95 % CI, 0.17–0.28), and 1.14 (95 % CI, 1.02–1.26) to 2.24 (95 % CI, 1.81–2.67), respectively. Overall, this synergistic effect exhibited a strengthening trend with increasing thresholds and longer durations (Fig. 4). This synergistic effect was further validated on the multiplicative scale (Fig. S4). Moreover, the synergistic effect was particularly evident in hypertensive heart disease, ischemic heart disease, acute myocardial infarction, hemorrhagic stroke, and chronic obstructive pulmonary disease (Table S6). Unexpectedly, under the HW95th-OZ95th definition, the estimated synergistic effect did not increase correspondingly with the intensification of heat wave and O_3 pollution levels; instead, it showed a slight decreasing gradient (Fig. 4).

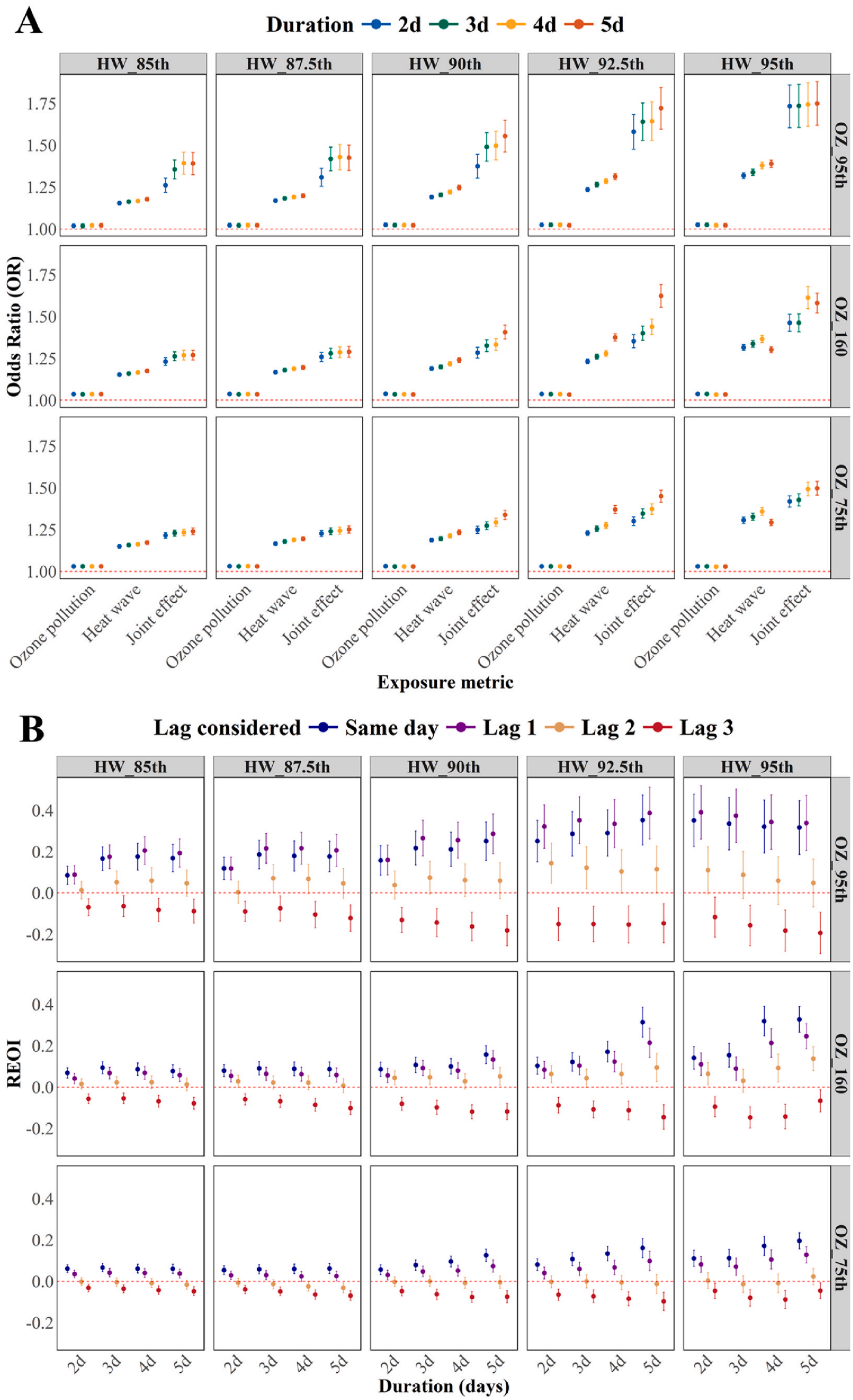

Fig. 3. Effect estimates by varying definitions of extreme events for cardiopulmonary mortality. (A) Associations between cardiopulmonary mortality and heat wave, O_3 pollution and joint of both of compound event. (B) Additive synergistic effect of heat wave and O_3 pollution summarized in relative excess risk due to interaction (REOI).

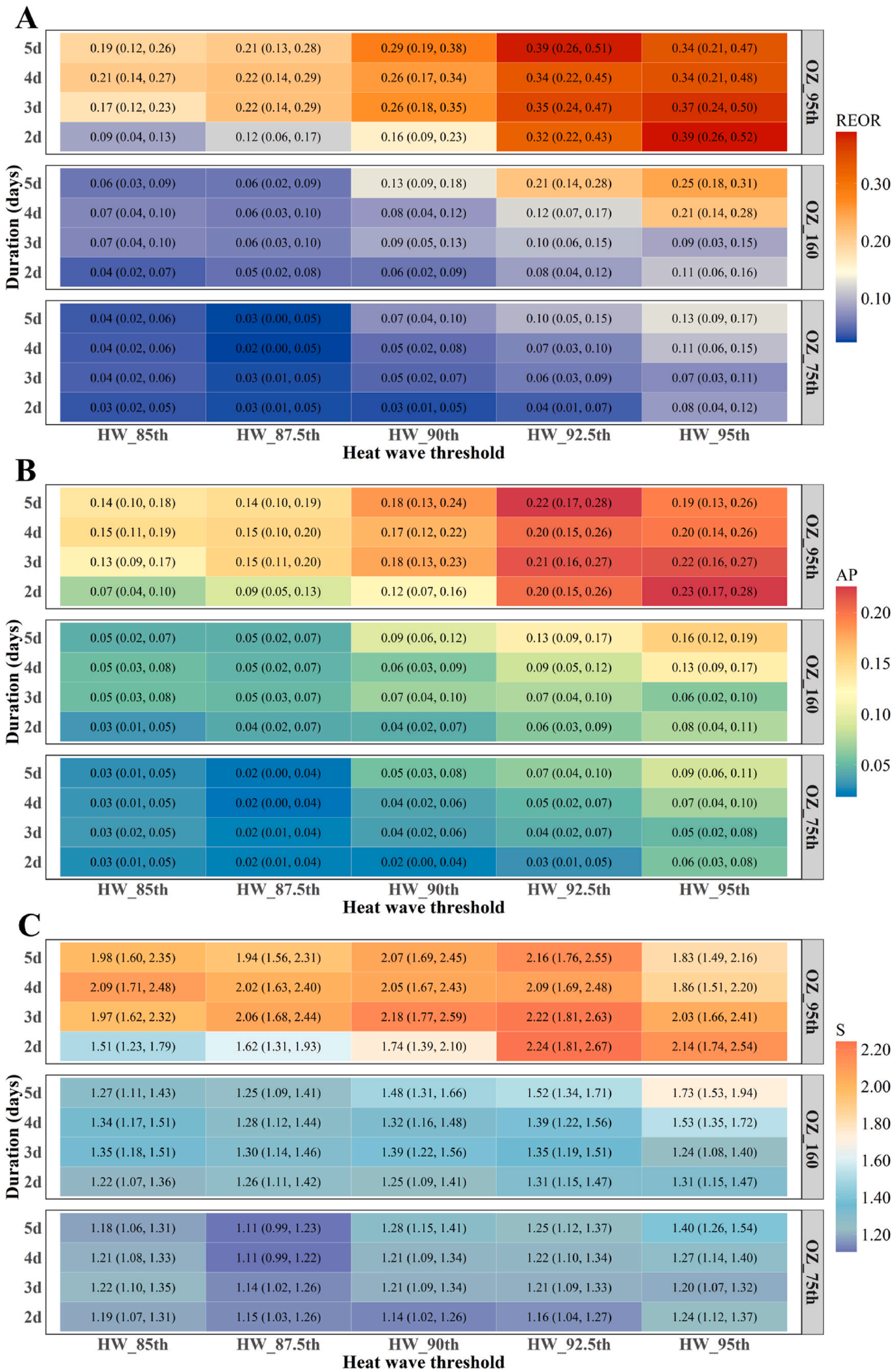

Fig. 4. Interactive effects of compound exposure to heat wave and O_3 pollution on cardiopulmonary mortality under 60 definitions.

3.5. Stratified analyses

Fig. 5 illustrates cardiopulmonary mortality risks associated with the three exposure events stratified by sex, age groups, and urban-suburban-rural areas. We observed that the independent effects of heat wave exposure on cardiopulmonary mortality were significantly higher in females than that in males, whereas the independent effects of exposure to O_3 were significantly higher in adults ≥65years than in adults <65 years ($P_{difference}$ < 0.05). However, no significant difference in independent effects of exposure to O_3 on cardiopulmonary mortality was detected across sex and age groups ($P_{difference}$ > 0.05). Although females and the adults ≥65years exposed to concurrent heat wave and O_3 pollution experienced higher cardiopulmonary mortality risks, this difference was not significant under medium- and high-levels of compound event exposures ($P_{difference}$ > 0.05). Across urban, suburban, and rural areas, the ORs for cardiopulmonary mortality associated with exposure to O_3 pollution-only events ranged from 1.01 (95 % CI, 0.97–1.04) to 1.05 (95 % CI, 1.04–1.06), showing no significant differences ($P_{difference}$ > 0.05). Nevertheless, the impact of heat wave-only and compound event exposures on cardiopulmonary mortality among residents in rural-dominated mixed region (MR) was significantly higher than in urban area (UR) ($P_{difference}$ < 0.05), with this difference being more pronounced during compound events. Notably, under the definition of HW95_5d-OZ95th, the OR for cardiopulmonary mortality associated with compound event exposure in rural area (RU) surged sharply to 2.77 (95 % CI, 1.61–3.92), highlighting the structural vulnerability of marginalized communities facing extreme hazards. It is important to acknowledge that the relatively wide confidence intervals associated with extreme compound events exposure (HW95th_5d-OZ95th), particularly within the RU subgroup, likely reflect limited sample sizes and increased uncertainty, which may consequently compromise the precision and reliability of these estimates.

3.6. Excess cardiopulmonary mortality

According to various definitions, the total excess cardiopulmonary deaths attributed to heat wave-only, O_3 pollution-only, and compound events exposure in Anhui during 2013–2021 were estimated to range from 4358 (95 % CI, 4122–4597) to 9873 (95 % CI, 9243–10,508), 555 (95 % CI, 254–859) to 4503 (95 % CI, 3586–5429), and 243 (95 % CI, 201–285) to 3227 (95 % CI, 2974–3480), respectively. As the definitions became more stringent, the related excess cardiopulmonary deaths gradually decreased. The proportion of excess deaths attributed to O_3 pollution-only and compound events exposure among the total excess

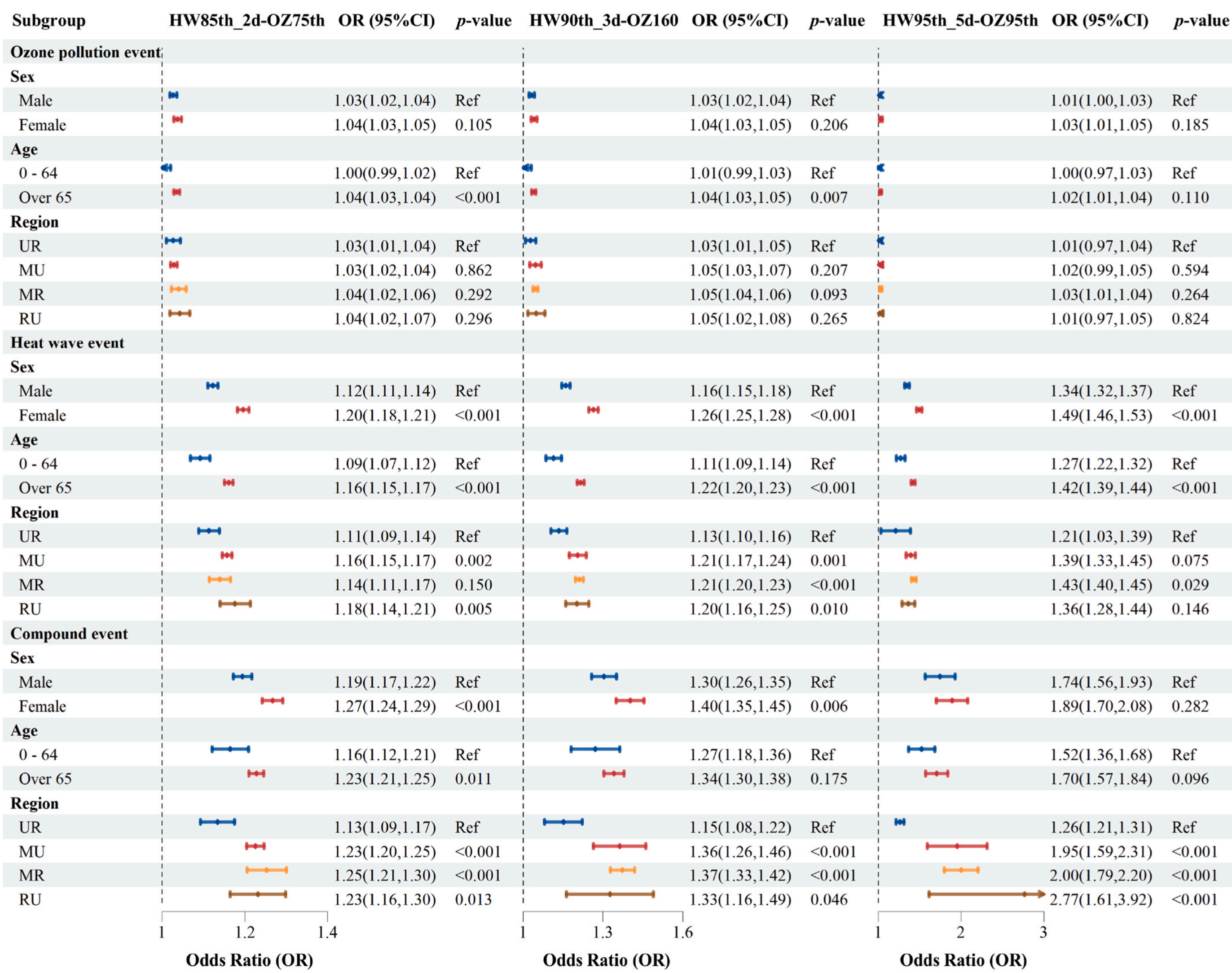

Fig. 5. Cardiopulmonary mortality risk associated with three extreme events by sex, age and region groups under various definitions. In the forest plots, colors indicate subgroups: blue represents males, adults <65 years, and UR; red represents females, adults ≥65 years, and MU; orange and brown correspond MR and RU, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

deaths attributable to the three extreme events also declined from 25.28 % to 18.33 %–10.76 % and 4.71 %, respectively. For heat wave events-only, however, this proportion increased from 56.09 % under the definition of HW85th_2d-OZ75th to 84.53 % under the definition of HW95th_5d-OZ95th (Fig. 6; Table S7). Furthermore, the temporal distribution of excess cardiopulmonary deaths attributed to the three extreme events exposure was uneven (Fig. S5). In 2013, the excess deaths attributed to heat wave-only events (1382 to 1438 deaths) accounted for over 99 % of the total excess deaths (1382 to 1446 deaths) attributable to the three extreme events. With the gradual intensification of O_3 pollution, the number of excess deaths attributed to the O_3 pollution-only and compound events sharply increased after 2017. Under various definitions, the proportion of excess deaths attributed to the compound events in the total excess cardiopulmonary deaths caused by three types of events ranged from 8.97 % (95 % CI, 7.92 %–10.00 %) to 21.43 % (95 % CI, 20.80 %–22.20 %) in 2017–2021, whereas this proportion ranged only from 0.29 % (95 % CI, 0.26 %–0.32 %) to 7.94 % (95 % CI: 7.98 %, 8.00 %) from 2013 to 2016. At the township level, the

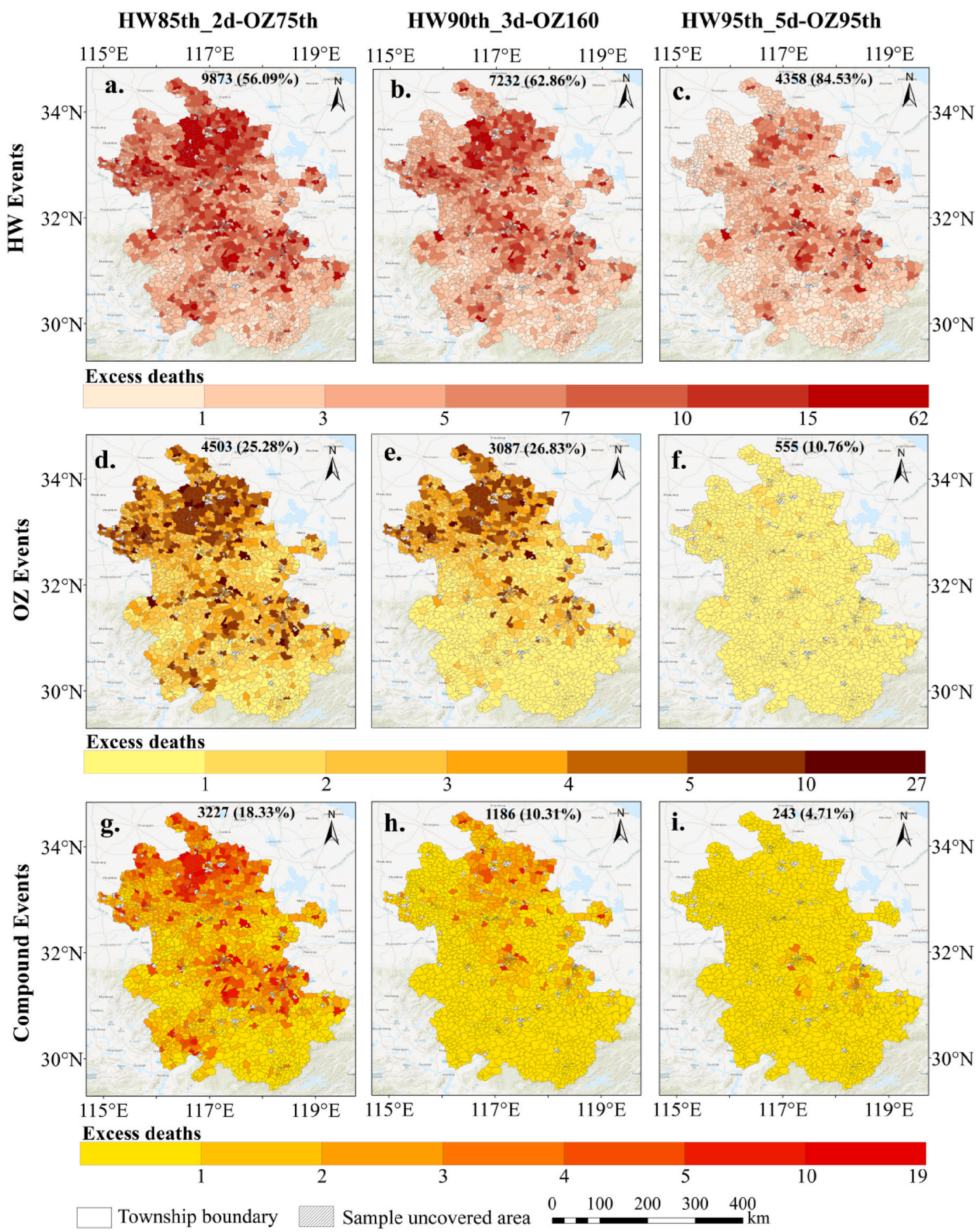

Fig. 6. Spatial distribution of attributable the total excess deaths attributed to three extreme events under various definitions in Anhui Province during the warm seasons from 2013 to 2021.

excess deaths attributed to the three extreme events were concentrated in densely populated northern and central regions; however, under exposure to more extreme compound event (HW95th_5d-OZ95th), only townships in the central-eastern region were observed to bear a disproportionately burden of excess cardiopulmonary mortality (Fig. 6i).

3.7. Sensitivity analysis

Sensitivity analyses showed that defining heat wave events using daily mean air temperature instead of apparent temperature yielded effect estimates and lag patterns similar to the main analysis (Table S8); after adjusting for NO_2 in the model, the independent and joint effects of the three extreme events exposure on cardiopulmonary mortality were slightly enhanced (Table S9); when adjusting for $PM_{2.5}$ and relative humidity, the corresponding effect estimates slightly decreased (Tables S10 and 11); effect estimates in the two-pollutant model exhibited minor fluctuations compared to the single-pollutant model, but the direction of the effect remained consistent (Table S12); recalculating cluster-robust standard errors by county led to a 12.61–25.18 % widening of confidence intervals; however, the direction and significance of the effects remained unchanged, indicating a limited influence of spatial autocorrelation (Tables S13 and 14); validation analyses based on subjects from 2013 to 2019 further confirmed the robustness of the study conclusions (Table S15).

4. Discussion

To the best of our knowledge, this is the first township-level space-time-stratified case-crossover study in China to comprehensively investigate the association and interactive effects of compound exposure to heat wave and O_3 pollution with cause-specific cardiopulmonary mortality. Based on a more refined and systematic assessment of 550,634 cardiopulmonary deaths in Anhui Province, China from 2013 to 2021, we found consistent evidence that exposure to both heat wave and O_3 pollution were significantly associated with increased odds of multi-cause cardiopulmonary deaths. Notably, the increased mortality risks associated with compound exposure to these two hazards were more pronounced for hypertensive heart disease, ischemic heart disease, acute myocardial infarction, hemorrhagic stroke, and chronic obstructive pulmonary disease deaths. Females, older adults (≥65years), and residents in MR were more vulnerable to heat wave and O_3 pollution exposure. From 2013 to 2021, we estimated that the total excess cardiopulmonary deaths attributable to compound exposure to heat wave and O_3 pollution ranged from 243 (95 % CI: 201–285) to 3227 (95 % CI: 2974–3480) under various definitions, predominantly concentrated in densely populated townships in northern and central-eastern regions.

By constructing 20 gradient heat wave definitions, we observed positive associations between heat wave exposure and mortality risks for all cardiopulmonary diseases included in the study, and these associations exhibited a gradually strengthening trend with heat wave intensity and duration. A multinational cohort study across 567 cities in 27 countries reported a pooled relative risk (RR) of 1.10 (95 % CI: 1.07–1.14) for cardiovascular mortality associated with extreme heat exposure (Alahmad et al., 2023), which was lower than ours (ORs ranging from 1.16 to 1.44). A recent nationwide time-series study conducted in China also consistently found more pronounced acute effects of heat wave events on circulatory mortality (Du et al., 2024). These associations may be related to heat-induced thermoregulatory dysfunction and heat stress exacerbating atherosclerosis (Ahmadnezhad et al., 2013; Zhu et al., 2025). Furthermore, a case-crossover study in Jiangsu Province, China, confirmed that cardiovascular mortality risk associated with heat wave exposure depends on its intensity and duration (Xu et al., 2023). Our findings are consistent with these studies to some extent, while the identified risk associations at lower thresholds (OZ85th and OZ87.5th) of heat wave exposure highlight the need for future focus on milder but more frequent heat events.

Similarly, our findings of significant associations between O_3 pollution exposure and cardiopulmonary mortality have been documented in previous studies. A 17-year prospective cohort study across six US states found a hazard ratio (HR) of 1.03 (95 % CI: 1.01–1.06) for cardiovascular mortality per 10 ppb increase in O_3 concentration (Lim et al., 2019). Another time-series study in 96 counties of Jiangsu Province, China, showed that for every 10 $\mu g/m^3$ increase in O_3 concentration, cardiovascular and respiratory mortality increased by 0.85 % (95 % CI: 0.75–0.96 %) and 1.15 % (95 % CI: 0.94–1.36 %), respectively (Guo et al., 2023). Moreover, we observed even more pronounced risk estimates between O_3 pollution exposure and chronic obstructive pulmonary disease (COPD) mortality. As a strong oxidant, O_3 can preferentially damage the respiratory system through oxidative stress and inflammatory pathways. It reacts with airway mucosa to generate reactive oxygen species (ROS), impairing alveolar epithelial barrier function, thereby exacerbating acute exacerbations of COPD (Kim et al., 2011; Niu et al., 2020). However, the mortality risk from cardiopulmonary diseases under high-level O_3 pollution exposure was relatively lower, which may be related to protective behaviors (e.g., reduced outdoor activities) on high pollution days and “survivor bias”. This bias occurs when the most vulnerable individuals have already died during periods of lower O_3 exposure, resulting in a more resilient population at higher exposure levels and potentially causing an underestimation of the true health risks associated with high-level O_3 pollution (Miller et al., 2012). Thus, accounting for the potential impact of survivor bias is crucial when interpreting the exposure-response relationship at high exposure levels.

Compound exposure to heat wave and O_3 pollution is associated with a higher risk of cardiopulmonary mortality, with synergistic amplification effects being more pronounced in sensitive diseases. Epidemiological evidence reveals mutual modification effects between these two extreme events, further exacerbating exposure risks. For instance, research across eight European cities observed that per 10 $\mu g/m^3$ increase of O_3 concentration was associated with an increased mortality risk of 0.44 % (95 % CI: −0.05 %–0.93 %) and 0.54 % (95 % CI: 0.06 %–1.02 %) at low (<25th percentile) and high temperatures (>75th percentile), respectively. The increase in cardiovascular mortality risk caused by O_3 pollution under high-temperature conditions was significantly greater than that under low-temperature conditions (Chen et al., 2018). Qi et al. (2023) provided further epidemiological evidence showing that O_3 pollution exacerbated cerebrovascular mortality risks related to high-temperature weather in 250 Chinese cities. Similarly, a time-series study in Nanjing, China, revealed that higher temperatures and O_3 concentrations can increase ischemic heart disease mortality via mediating effects (Gong et al., 2024).

From a molecular and physiological perspective, the synergistic effects of heat wave and O_3 pollution on cardiopulmonary health may be explained by several interrelated biological mechanisms. Both heat stress and O_3 exposure independently induce systemic oxidative stress and inflammation, which may be amplified when combined, leading to enhanced endothelial dysfunction and vascular inflammation—key processes in cardiovascular and respiratory pathogenesis (Cui et al., 2013; Arjomandi et al., 2015).Additionally, elevated temperatures can promote hypercoagulability and alter blood viscosity, thereby intensifying the pro-thrombotic effects of short-term O_3 exposure, increasing the risk of ischemic events such as myocardial infarction and ischemic stroke (Niu et al., 2022; Wang et al., 2025c). This mechanistic hypothesis has been validated in a study examining the association between heat wave and O_3 pollution exposure and ischemic stroke mortality in Jiangsu Province (Wang et al., 2025d). Heat wave may also disrupt autonomic nervous system regulation and impair cardiopulmonary function, which, together with O_3-induced respiratory tract irritation and decreased lung function, further heighten susceptibility to adverse outcomes (Kahle et al., 2015; Roths et al., 2024). Comound exposure may overwhelm systemic compensatory and repair mechanisms, contributing to organ dysfunction and exacerbating pre-existing cardiopulmonary diseases (Alahmad et al., 2023; Du et al., 2024). Furthermore, cardiopulmonary diseases with higher susceptibili ty to compound exposure to heat wave and O_3 pollution—including hypertensive heart disease, acute myocardial infarction, chronic ischemic heart disease, and chronic obstructive pulmonary disease, have all been confirmed in single-exposure studies (Lim et al., 2019; Chen et al., 2023; Zhu et al., 2024). Nonetheless, the intricate interplay among pathological pathways, organ interactions, and systemic compensatory imbalances induced by combined heat and O_3 exposure remains incompletely understood. Therefore, future research is warranted to explore these refined mechanisms, which will be crucial for developing targeted public health strategies aimed at mitigating the compounded risks posed by extreme heat and air pollution events.

The synergistic amplification effects observed under extreme compound events require cautious interpretation. Using the HW95th-OZ95th definition, our study identified a gradient-decreasing trend in synergistic effects, a phenomenon similarly reported by Du et al. (2024). Beyond the reduced statistical power caused by smaller sample sizes, this phenomenon may relate to physiological adaptation thresholds and competing risk bias. Specifically, individuals exposed to extremely high levels of heat wave or O_3 pollution might reach physiological adaptation limits (Miller et al., 2012), which could attenuate the magnitude of further synergistic effects. Meanwhile, competing risk bias might arise because extreme independent events (Van-Walraven and Mcalister, 2016) such as severe heat wave or intense O_3 pollution—could precipitate earlier cardiopulmonary deaths, thereby masking the synergistic effects of compound exposures. Notably, the complex interactions among pollutants and the bidirectional modulation by meteorological factors can affect the estimation of synergistic effects to some extent. After adjusting for NO_2, the negative association between NO_2 and O_3 formation (e.g., photochemical titration effect) was controlled, thereby unmasking the true synergistic effect of heat wave and O_3 exposure, with slightly amplified effect estimates. Conversely, when including $PM_{2.5}$ and relative humidity in the model, the negative confounding between $PM_{2.5}$ and heat wave (where high temperature promotes dispersion, reducing $PM_{2.5}$ concentration) and the bidirectional modulation by relative humidity (which enhances heat stress but inhibiting O_3 formation) were corrected. This adjustment avoided overestimation caused by ignoring competing risks or collinearity, resulting in a more conservative estimate of synergistic effects. These findings underscore the critical importance of adequate confounding control in environmental mixture analyses to ensure accurate risk estimation.

Females, older adults, and residents in marginal transitional zones experienced stronger synergistic effects from heat wave and O_3 pollution. Physiologically, females exhibit heightened inflammatory responses and distinct immune regulation within cardiopulmonary pathways (Cabello et al., 2015; Mishra et al., 2016), rendering them more sensitive to these two hazards. Additionally, women's family caregiving roles often lead to more frequent exposure to both indoor and outdoor heat wave and O_3 pollution, thereby increasing their cumulative exposure risk (Van-Steen et al., 2019; Liu et al., 2021). Older adults demonstrate reduced physiological functions, particularly diminished cardiopulmonary reserve, impaired thermoregulation, and cognitive dysfunction (Shumake et al., 2013; Xi et al., 2024), which collectively lower their tolerance to heat wave and O_3 pollution. Social isolation, burden of chronic disease, and limited access to healthcare services further exacerbate their vulnerability (Kenney et al., 2014; Sun et al., 2023). Consistent with these factors, the predominance of older adults, low educational attainment, and farming occupations within the study population likely reflects limited adaptive capacity (e.g., restricted access to air conditioning, healthcare services, and health risk information), which may contribute to their heightened vulnerability and help explain the observed strong associations between heat wave and O_3 pollution exposure and increased cardiopulmonary mortality. When considering a more refined urban-rural fourfold structure, marginalized transitional zones showed the most pronounced vulnerability to compound events exposure. This heightened risk may be linked to the loss of ecological buffers caused by rapid urbanization, alongside simultaneous increases in population density and pollution levels (Li et al., 2022; He, 2023; Yang et al., 2022). The identification of this novel vulnerability hotspot highlights the limitations inherent in the traditional urban-rural dual structure and suggests that future environmental health equity research should adopt more nuanced spatial frameworks for analysis. However, due to the absence of direct behavioral, occupational, or physiological data, the inferred population vulnerability patterns still require further validation by incorporating relevant measurements.

Significant spatiotemporal heterogeneity has been observed in excess mortality attributed to heat wave and O_3 pollution exposure. The 2013 heat wave predominantly drove early excess mortality; however, with ongoing climate warming and intensified regional photochemical pollution, deaths related to O_3 pollution and compound events have increased exponentially since 2017. Beyond the direct effects of rising temperature and increased O_3 exposure, Anhui has experienced a significant demographic shift characterized by accelerated population aging, with the proportion of elderly individuals rising from 12.30 % to 15.44 % between 2017 and 2021 (https://tjj.ah.gov.cn/). This demographic transition has substantially heightened intrinsic vulnerability, particularly among older adults who exhibited diminished physiological resilience to environmental stressors (Wu et al., 2023; Xi et al., 2024). Concurrently, rapid urbanization advanced from 53.49 % to 59.39 %, leading to a spatial concentration of populations in areas with high O_3 exposure, where urban heat island effects and localized air pollution synergistically exacerbate health risks (Sun et al., 2023). Despite overall improvements in healthcare infrastructure, the uneven distribution of healthcare resources has limited timely access to critical interventions during extreme environmental events (Yan et al., 2022), disproportionately increasing mortality risk in underserved regions. Under the combined influence of these factors, excess mortality related to heat wave and O_3 pollution exhibited distinct spatiotemporal clustering patterns. Notably, while heat wave remain the primary cause of deaths, the accelerated rise in excess mortality from compound events suggests that future compound climate hazards may emerge as major public health threats. As definitions of extreme events become more stringent, high-intensity and long-duration heat wave increasingly contribute to excess cardiopulmonary mortality. To counter this escalating public health threat, integrated interventions must target both exposures simultaneously. These include establishing dynamic joint early warning systems to trigger public protections, coupled with pre-emptive emission controls during heat episodes to suppress O_3 formation. Improving urban planning by increasing green spaces, enhancing ventilation corridors, and optimizing building materials and layouts to reduce heat accumulation, as well as implementing prioritized safeguards for vulnerable populations, are equally critical to reduce compound risks.

Several strengths of this study should be noted. First, the nine-year investigation encompassed over 0.5 million cardiopulmonary deaths across more than ten specific causes. The large and representative sample size provided robust statistical power, clarified susceptible disease categories and expanded the current epidemiological evidence base. Second, heatwave events were defined using apparent temperature, and 60 gradient compound exposure scenarios were constructed. These diverse definitions of compound events not only enabled accurate analysis of dynamic changes in cardiopulmonary mortality along the exposure gradient but also revealed nonlinear variations in synergistic effects under more extreme conditions. This approach facilitates a more comprehensive understanding of the impact of compound heat wave and O_3 pollution exposure on cardiopulmonary mortality. Finally, the study assessed urban–suburban–rural disparities in compound exposure risk within a refined urban–rural fourfold structure, identifying the MR as a novel vulnerable hotspot.

We acknowledge several limitations in this study. First, exposure data were derived from validated gridded datasets rather than individual-level measurements, which inevitably introduces non-differential exposure misclassification that may bias effect estimates toward the null by attenuating the true exposure–response relationship. Therefore, our findings likely represent conservative estimates of the true associations. Second, although rigorous quality control procedures were applied during data extraction, the large sample size and inclusion of multiple disease categories mean that the possibility of ICD coding errors cannot be completely ruled out, and their potential impact on study results remains difficult to quantify. Third, due to relatively low daily death counts from hypertensive heart disease and chronic obstructive pulmonary disease, coupled with limited statistical power, subgroup analyses by population for different cardiopulmonary disease subtypes were not conducted, thereby restricting in-depth interpretation of disease-specific effects. Fourth, while the space-time-stratified case-crossover design effectively controls for time-invariant confounders and spatial effects, residual confounders that may vary for an individual within the case and control days (e.g., unmeasured acute health events or personal protective behaviors) cannot not be excluded and might introduce bias into the risk estimates. Finally, our study was conducted in one province in China, caution is warranted when extrapolating the findings to other regions and countries due to potential heterogeneity in climatic conditions, population vulnerability pollution profiles.

5. Conclusion

In conclusion, this township-level space-time-stratified case-crossover study provided consistent evidence that heat wave interacts synergistically with O_3 pollution to trigger multi-cause cardiopulmonary deaths, especially for hypertensive heart disease, acute myocardial infarction, ischemic heart disease, hemorrhagic stroke, and chronic obstructive pulmonary disease. As exposure levels to heat wave and O_3 pollution increase, females, older adults, and residents living in the rural-dominated mixed region exhibited progressively greater vulnerability to these concurrent hazards. In the context of intensifying climate warming, this study offers critical scientific evidence to support enhanced comprehensive surveillance of cardiopulmonary diseases sensitive to heat wave and O_3 pollution exposure, to identify vulnerable populations and regions, and to establish joint early warning systems.

CRediT authorship contribution statement

Kang Ma: Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Fengman Fang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yuesheng Lin:** Writing – review & editing, Validation, Supervision. **Fei Tong:** Visualization, Formal analysis, Data curation. **Cheng He:** Writing – review & editing, Supervision, Methodology. **Youru Yao:** Writing – review & editing, Supervision, Investigation. **Jingli Zhu:** Validation, Resources. **Huadong Wang:** Validation, Resources. **Xiuya Xing:** Supervision, Resources, Methodology. **Feiyan Zhang:** Formal analysis. **Ruoxi Li:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2025.122635.

Data availability

Data will be made available on request.

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